

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 418 — Past Questions & Materials
by Renance

Course: PHY 418
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — PHY 418 Nuclear Physics

Level 400 • First Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria
College of Science
Department of Physics



Academic Session: 2023/2024
Course Title: NUCLEAR PHYSICS

Semester: 1ST
Date: 13/5/2024

Course Code: PHY 418

Time Allowed: 2 hrs.

Instructions: ANSWER ANY THREE QUESTIONS

Question 1

- a) Discuss briefly the following fundamental forces of nature: i. Strong nuclear force, ii. Electromagnetic force iii. Weak force and iv. Gravitational force. (8marks)
- b) Discuss the following nuclear terminology: i. Atomic Mass Unit (amu), ii. Mass defect iii. Binding Energy (7marks)
- c) i) Given a detail explanation of the Odd-Even rule of nuclear stability. (3marks)
ii) Based on the even-odd rule, predict which one would you expect to be radioactive in pair? 1. ${}_8\text{O}^{16}$ & ${}_8\text{O}^{17}$ 2. ${}_{17}\text{Cl}^{35}$ & ${}_{17}\text{Cl}^{36}$ 3. ${}_{10}\text{Ne}^{20}$ & ${}_{10}\text{Ne}^{17}$ 4. ${}_{20}\text{Ca}^{40}$ & ${}_{20}\text{Ne}^{45}$
5. ${}_{80}\text{Hg}^{195}$ & ${}_{80}\text{Hg}^{196}$ (5marks)

Question 2

- a) i) What is radioactivity? Distinguish between natural and artificial radioactivity. (4marks)
ii) state five properties of radioactivity (4marks)
- b) i) What is half-life? Derive the equation for half-life of radioactive element which undergoes disintegration. (4marks)
ii) Derive the radioactive decay equation. (3marks)
- c) Write briefly on i. alpha decay and state three of its characteristics ii. beta decay and state three of its characteristics and iii. gamma decay and state three of its characteristics (8marks)

Question 3

- a) i) What is nuclear model? (2marks)
ii) Discuss the liquid drop model and state four of the justifications. (5marks)
- b) i) Bethe-Weizacker Semi-empirical Mass Formula (SEMF) uses the five factors that contribute to the binding energy of nuclei to predict the binding energy of a nucleus. State these five factors and their equations. (5marks)

Photons

- ii) From the equation of the five factors, state the Bethe-Weizacker Semi-empirical Mass Formula (SEMF). (3marks)
- c) The atomic mass of Zn isotope $_{30}\text{Zn}^{64}$ is 63.920 amu. Compare the binding energy from the nuclear composition and predicted by SEMF (given $M_p=1.0072$, $M_n=1.0086$) (8marks)

Question 4

- a) i) Explain the shell model of the nucleus (3marks) ✓
- ii) Outline some of the drawbacks of the liquid drop model of the atom which the Shell model of the nucleus addressed (4marks)
- b) i) Tabulate the similarities between Shell model and the Hartree-Fock theory (4marks) ✓
- ii) Tabulate the differences between Shell model and the Hartree-Fock theory (4marks) ✓
- c) i) Show that the force binding the nucleons within the nucleus cannot possibly be gravitational, if the diameter of the nucleon is 10^{-15}m and the distance between two interacting nucleons is $2 \times 10^{-15}\text{m}$. (8marks)

Question 5

- a) i) What is Q-value in nuclear reaction. (3marks)
- ii) What is Nuclear Fission. (2marks)
- ii) Discuss controlled thermonuclear fission. (3marks)
- b) i) What is "effective neutron multiplication factor"? (3marks)
- ii) State the conditions for the following nuclear chain reaction to occur; 1. Subcritical 2. Critical 3. Supercritical. (6marks)
- c) i) Differentiate between Research reactor and Power reactor. (3marks)
- ii) With the aid of a labelled diagram, identify the various component of the nuclear reactor. (4marks)

Paper 2 of 2 — Nuclear physics PHY 418 pq

Level 400 • First Semester • PastQuestion

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- ii) Based on the even-odd rule, predict which one would you expect to be radioactive in pair? 1, ^{101}Po & ^{201}Po 2. ^{135}C & ^{137}C 3. ^{238}U & ^{235}U 4. ^{40}Ca & ^{40}Ar 5. ^{200}Hg & ^{200}Au (Smarks) .

Question 2

- a) i) What is radioactivity? Distinguish between natural and artificial radioactivity. (4marks)

ii) state five properties of radioactivity (4marks)

b) i) What is half-life? Derive the equation-for half-life of radioactive element which undergoes disintegration. (4marks)

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